WORKED SOLUTION

STEP III 1990, Question 7

Pure • Curve sketching

STEP III 1990, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q7

The key idea

Implicit differentiation, done twice, gives $\mathrm{d}y/\mathrm{d}x$ and $\mathrm{d}^2y/\mathrm{d}x^2$ at each point, and these are exactly the coefficients in the Taylor series. For the inequality, the trick is to notice that the equation is only a *quadratic* in y: a point (x, y) can exist only if that quadratic has real roots, so its discriminant is non-negative. Solving the quadratic explicitly then shows the whole curve, two branches that meet where the discriminant vanishes.

Solution

Throughout, write $F(x, y) = xy^2 + x^3 + a^2y - a^3$. Substituting shows that $F = 0$ at P, Q and R; for instance at R we get $a \cdot a^2 + a^3 - a^3 - a^3 = 0$.

The Taylor series at P, Q and R

Differentiating $F(x, y) = 0$ with respect to x, treating y as a function of x,

$$y^2 + 2xyy' + 3x^2 + a^2y' = 0, \qquad \text{so} \qquad y' = -\frac{y^2 + 3x^2}{2xy + a^2}.$$

Differentiating the first equation again (the term $2xyy'$ gives $2yy' + 2xy'^2 + 2xyy''$):

$$4yy' + 2xy'^2 + 6x + (2xy + a^2)\, y'' = 0.$$

At each point the denominator $2xy + a^2$ is non-zero, so these equations determine y' and y'', and then $y = y_0 + y'h + \frac{1}{2}y''h^2 + \cdots$.

At $P(0, a)$: $y' = -a^2/a^2 = -1$. Then $4a(-1) + a^2y'' = 0$, so $y'' = 4/a$ and

$$y = a - h + \frac{2}{a}h^2 + \cdots.$$

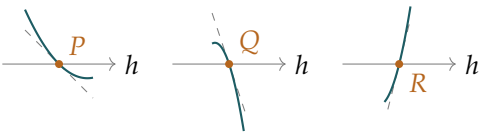
At $Q(a, 0)$: $y' = -3a^2/a^2 = -3$. Then $0 + 2a(9) + 6a + a^2y'' = 0$, so $y'' = -24/a$ and

$$y = -3h - \frac{12}{a}h^2 + \cdots.$$

At $R(a,-a)$: here $2xy + a^2 = -a^2$, so $y' = -4a^2/(-a^2) = 4$. Then $4(-a)(4) + 2a(16) + 6a - a^2 y'' = 0$, that is $22a = a^2 y''$, so $y'' = 22/a$ and

$$y = -a + 4h + \frac{11}{a}h^2 + \cdots .$$

So near P the curve is falling with gradient -1 and is concave up (bending above its tangent); near Q it falls steeply with gradient -3 and is concave down; near R it rises with gradient 4 and is concave up.



(Local shapes with $a = 1$; the dashed lines are the tangents.)

The inequality

Write the equation of C as a quadratic in y:

$$x\,y^2 + a^2\,y + (x^3 - a^3) = 0.$$

If $x \neq 0$ and (x,y) lies on C, this quadratic has the real root y, so its discriminant is non-negative:

$$a^4 - 4x(x^3 - a^3) \geq 0 \quad \Longleftrightarrow \quad 4x^4 - 4a^3x - a^4 \leq 0.$$

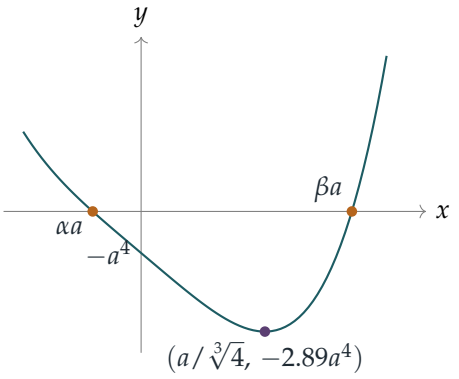
If $x = 0$ the left-hand side is $-a^4 < 0$, so the inequality holds there too. This proves the result.

The graph of $y = 4x^4 - 4a^3x - a^4$

The printed question has $4x^4 - 4a^3 - a^4$, which is surely a misprint for the quartic $g(x) = 4x^4 - 4a^3x - a^4$ we have just met. We have $g(0) = -a^4$ and $g'(x) = 16x^3 - 4a^3$, which vanishes only at $x_0 = a/\sqrt[3]{4} \approx 0.63a$, where g has its minimum. Using $x_0^3 = a^3/4$,

$$g(x_0) = a^3x_0 - 4a^3x_0 - a^4 = -a^4(1 + 3\cdot 4^{-1/3}) \approx -2.89a^4.$$

Since $g \to +\infty$ as $x \to \pm\infty$ and g is decreasing then increasing, it has exactly two zeros, $x = \alpha a$ and $x = \beta a$, where $\alpha \approx -0.246$ and $\beta \approx 1.072$ are the roots of $4t^4 - 4t - 1 = 0$ (there's no neat closed form, and we don't need one). Also $g''(x) = 48x^2 \geq 0$, so the graph is convex.



(Drawn with $a = 1$.) So every point of C has $\alpha a \leq x \leq \beta a$: the curve lives in a vertical strip.

The curve C

For $x \neq 0$, solving the quadratic gives the two branches

$$y_{\pm} = \frac{-a^2 \pm \sqrt{D}}{2x}, \qquad D = a^4 + 4a^3x - 4x^4,$$

which are real exactly on the strip and meet where $D = 0$, at $x = \alpha a$ and $x = \beta a$. There $y = -a^2/(2x)$, and the tangent is vertical, since that is where the denominator $2xy + a^2$ of y' vanishes. This gives the points $(\beta a, -0.47a)$ and $(\alpha a, 2.03a)$.

? **Ask yourself** What happens to the two branches as $x \to 0$?

Multiplying top and bottom by $\sqrt{D} + a^2$, and using $D - a^4 = 4x(a^3 - x^3)$, we get

$$y_+ = \frac{2(a^3 - x^3)}{a^2 + \sqrt{D}},$$

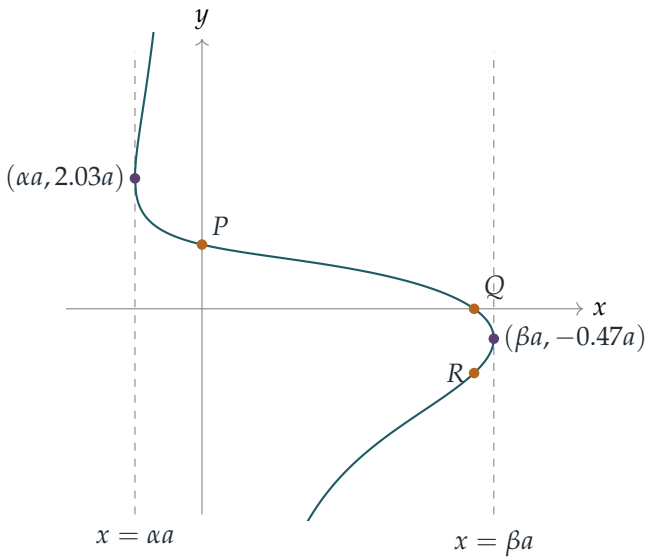
which is perfectly well behaved at $x = 0$ and equals a there. So y_+ is a single smooth branch through P. The other branch satisfies $y_- \approx -a^2/x$ for small x, so $y_- \to -\infty$ as $x \to 0^+$ and $y_- \to +\infty$ as $x \to 0^-$. This is the asymptote along the y-axis.

Some more facts help to pin down the sketch.

- There are no horizontal tangents: $y' = 0$ would need $y^2 + 3x^2 = 0$, that is $(0, 0)$, which is not on C.
- The curve meets $y = 0$ only where $x^3 = a^3$, at Q, and meets $x = 0$ only at P.
- At $x = a$, $D = a^4$, so $y_+ = 0$ (the point Q) and $y_- = -a$ (the point R).

Putting it together, C is one connected curve. It comes down from $+\infty$ just to the left of the y-axis, along y_-, to the vertical tangent at $(\alpha a, 2.03a)$. It turns back along y_+ through P (gradient -1), falls through Q (gradient -3) to the vertical tangent at

$(\beta a, -0.47a)$, then turns back along y_- through R (gradient 4) and goes down to $-\infty$ just to the right of the y-axis. It is concave up at P and concave down at Q, so there is a point of inflection between them.



(Drawn with $a = 1$; the dashed lines bound the strip found above.)

Checks

- With $a = 1.7$ we solved the quadratic for y at $x = x_0 + h$ near each point. The error in each two-term Taylor series fell by a factor of about 8 each time h was halved, which is the h^3 behaviour we'd expect.
- Taking 10^5 random values of x, the quadratic had real roots exactly when $4x^4 - 4a^3x - a^4 \le 0$.
- Bisection gives $\alpha \approx -0.2463$ and $\beta \approx 1.0724$. At $x = \pm 0.001$ (with $a = 1.7$), one root is close to a and the other close to $-a^2/x \approx \mp 2890$.

Common slips

- Forgetting the factor $\frac{1}{2}$: the h^2 coefficient is $\frac{1}{2}y''$, not y''.
- Dividing by x to use the discriminant without checking $x = 0$ separately.
- Missing that y_+ passes smoothly through P. Written as $(-a^2 + \sqrt{D})/(2x)$ it looks singular at $x = 0$, but it isn't.

Beyond STEP

Here we found the curve's behaviour near a point from a Taylor expansion of an implicitly defined function. The theorem that guarantees y is a smooth function of x wherever $\partial F/\partial y \neq 0$ is the *implicit function theorem*, a key result of multivariable analysis. The vertical tangents at the ends of the strip are exactly the points where $\partial F/\partial y = 2xy + a^2$ vanishes. This curve is a cubic in the plane, and the study of such curves leads on to algebraic geometry and elliptic curves. For more problems in this style, try Siklos, *Advanced Problems in Mathematics*.

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